

SKYLETIC

AI Marketing, Lead & Conversion Intelligence Platform

Build Specification for Google Antigravity / Claude Code

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1. Product Definition

Skyletic is a multi-tenant SaaS platform that connects Instagram, Facebook, YouTube, Google Ads, WhatsApp and a booking calendar into one AI-powered business intelligence portal.

Skyletic collects and normalizes marketing, content, conversation, lead and booking data; calculates reliable metrics; and uses AI to produce daily, weekly and monthly insights, recommendations and lead intelligence.

Social / Ads → Content interaction → Lead → WhatsApp conversation → Booking → Conversion

The product must answer: What happened? Why did it happen? Which channel/content generated leads? Which leads need attention? What should the business do next?

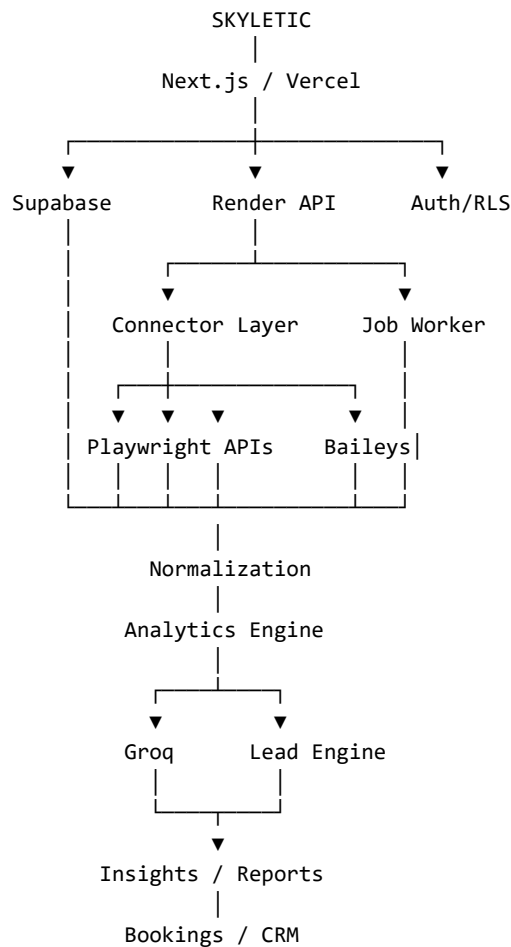
2. Core Goals

- One portal for Instagram, Facebook, YouTube and Google Ads performance.
- Use Playwright/Playwright MCP as a replaceable browser connector where appropriate and permitted.
- Connect the existing WhatsApp Baileys bridge and display conversations and lead status.
- Provide a booking calendar and connect bookings to leads and marketing sources.
- Provide daily, weekly and monthly dashboards with period comparisons.
- Use Groq for structured AI classification, explanations and recommendations.
- Track source attribution from content/campaign → lead → WhatsApp → booking.
- Build secure multi-tenant SaaS architecture.
- Do not make the product dependent on one scraping/browser method.

3. Recommended Stack

- Frontend: Next.js + TypeScript + Tailwind + shadcn/ui on Vercel.
- Database/Auth: Supabase PostgreSQL + Supabase Auth + Row Level Security.
- Backend/Workers: Render for long-running jobs and connector orchestration.
- Browser automation: Playwright MCP / Playwright on Render.
- AI: Groq behind an AI provider interface.
- WhatsApp: existing Baileys bridge through a secure backend adapter.
- Validation: Zod.
- Initial queue: PostgreSQL-backed jobs; add Redis/BullMQ only when needed.

4. High-Level Architecture



5. Main Modules

Overview:

Executive dashboard: KPIs, trends, AI summary, alerts and recommendations.

Social Analytics:

Instagram/Facebook/YouTube performance, content metrics, engagement and growth.

Google Ads:

Campaign, spend, impressions, clicks, CTR, CPC, conversions and CPL.

Content:

Unified content list and content-detail performance.

Leads:

Lead score, source, status, owner, conversation, booking and conversion.

WhatsApp:

Conversations, daily activity, lead status, unread/follow-up indicators.

Bookings:

Calendar, appointment status, source attribution and outcomes.

AI Insights:

Daily, weekly and monthly AI intelligence.

Reports:

Management reports and exports.

Connections:

Account connection, health and sync status.

Settings:

Organization, users, roles, AI and audit settings.

6. Customer Flow

1. Create account and organization.
2. Connect Instagram, Facebook, YouTube, Google Ads, WhatsApp and Calendar.
3. Verify connection status.
4. Run initial synchronization.
5. Normalize incoming data.
6. Calculate metrics in SQL/TypeScript.
7. Classify comments/conversations with Groq.
8. Create/update leads and attribution.
9. Create/update bookings.
10. Generate daily, weekly and monthly insights.
11. Display recommendations and follow-up actions.

7. Social Connector Architecture

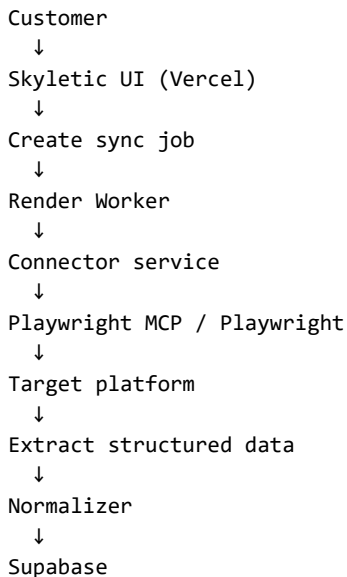
```
interface SocialConnector {  
  connect(accountId: string): Promise<void>;  
  disconnect(accountId: string): Promise<void>;  
  getStatus(accountId: string): Promise<ConnectionStatus>;  
  syncProfile(accountId: string): Promise<void>;  
  syncContent(accountId: string): Promise<void>;  
  syncComments(accountId: string): Promise<void>;  
  syncMetrics(accountId: string): Promise<void>;  
}
```

Initial targets: Instagram, Facebook and YouTube. A connector may use an official API/OAuth flow, Playwright/browser automation or another permitted source. The core application must not depend on the acquisition method.

- Long-running browser work runs on Render, not Vercel.
- Protect MCP/browser endpoints with authentication.

- Never expose MCP tokens or connector secrets to the frontend.
- Use isolated sessions per customer/account as the product scales.
- Handle CAPTCHA/MFA as a human-action-required state; never attempt bypass.
- Store sync status, last_sync_at, errors and source metadata.
- Prefer official APIs/OAuth where they provide the required capability and permitted commercial use.

8. Playwright MCP / Render Flow



Do not build the product as a scraper. Playwright is only one interchangeable data-acquisition layer.

9. Google Ads

Use the official Google Ads API/OAuth for production wherever practical. Browser automation is not the default when an official API provides the required data.

Target: campaign/ad group/ad → spend, impressions, clicks, CTR, CPC, conversions, conversion value and cost per conversion.

Support attribution using UTM parameters, campaign identifiers, landing-page identifiers and reliable lead-source metadata.

10. WhatsApp / Baileys

Skyletic connects to the existing Baileys bridge through a secure backend adapter.

Baileys → Skyletic
 conversation created/updated
 message received/sent
 contact identified
 lead created/updated
 lead status changed
 unread count
 last activity
 appointment intent
 follow-up required

Never expose bridge credentials to the frontend. For commercial scale, evaluate the official WhatsApp Business Platform/Cloud API and applicable policies.

11. Booking Calendar

- Create/update/cancel booking.
- Lead/customer association.
- Assigned salesperson.
- Date, time and timezone.
- Status: Scheduled, Confirmed, Completed, No-show, Cancelled.
- Marketing source/campaign/content attribution.
- Notes and outcome.
- Reminder/follow-up status.

12. Attribution

Instagram Reel
↓
Comment / interaction
↓
WhatsApp conversation
↓
Lead
↓
Booking
↓
Won

Lead should retain first-touch source, latest-touch source, platform, content/campaign IDs, WhatsApp conversation, booking and attribution confidence. Never claim deterministic attribution without evidence.

13. Lead Intelligence

Signals:

purchase intent
pricing question
availability question
payment-plan question
booking request
callback request
product/service question
complaint/support
spam/irrelevant
urgency
recency
prior engagement

```
{  
  "intent": "purchase",  
  "topic": "pricing",  
  "sentiment": "positive",  
  "lead_score": 94,  
  "priority": "high",  
  "requires_followup": true,  
}
```

```
"reason": "Customer asked for price and payment-plan information."
}
```

Lead score is a prioritization signal, not a guarantee of conversion.

14. AI Architecture

Application code calculates authoritative numbers. Groq explains verified evidence.

Application calculates:
views, likes, shares, engagement rate
growth %, CPL, CPA
lead counts and funnel conversion
rankings and period comparisons

Groq explains:
important changes
supported contributing factors
customer intent
content themes
risks
opportunities
recommended actions

All structured AI output must be validated with Zod. Store model, prompt version, reporting period and evidence.

15. Daily / Weekly / Monthly Intelligence

DAILY

Yesterday vs comparable prior period
Top/bottom content
New leads
WhatsApp activity
Ad performance
Bookings
Alerts
Recommended actions

WEEKLY

This week vs last week
Platform performance
Content patterns
Lead sources
Ad efficiency
WhatsApp conversion
Booking conversion
Recommendations

MONTHLY

Executive summary
Social performance
Ads
Leads
WhatsApp
Bookings
Attribution
Anomalies

Customer intent themes
Next-month plan

16. Database Schema

organizations
organization_members
social_accounts
content
content_metrics
comments
leads
lead_events
ad_accounts
ad_campaigns
ad_metrics
whatsapp_conversations
whatsapp_messages
bookings
ai_insights
sync_jobs
audit_logs

Every tenant-owned table must have `organization_id` and RLS. Use UUIDs, timestamps, foreign keys, unique external IDs and indexes for tenant/date/filter queries.

17. Suggested Key Fields

organizations:
`id, name, slug, timezone, created_at`

social_accounts:
`id, organization_id, platform, external_id, account_name, status, credential_reference, last_sync_at, sync_error`

content:
`id, organization_id, social_account_id, platform, external_id, content_type, title, text, url, published_at`

content_metrics:
`content_id, views, likes, comments, shares, saves, reach, impressions, captured_at`

comments:
`id, organization_id, content_id, external_id, author_name, text, published_at, sentiment, intent, lead_score, ai_reason`

leads:
`id, organization_id, contact_name, phone, email, source_platform, source_content_id, campaign_id, first_touch_source, latest_touch_source, status, lead_score, owner_user_id, whatsapp_conversation_id, booking_id, created_at, updated_at`

whatsapp_conversations:
`id, organization_id, external_id, contact_name, phone, status, assigned_user_id, first_message_at, last_message_at, lead_id`

```

whatsapp_messages:
id, organization_id, conversation_id, external_id,
direction, message_type, text, sent_at

bookings:
id, organization_id, lead_id, title, start_at, end_at,
timezone, status, owner_user_id, source, notes, outcome

ai_insights:
id, organization_id, insight_type, period_start, period_end,
summary, evidence, recommendations, model, created_at

sync_jobs:
id, organization_id, connector, job_type, status,
started_at, completed_at, error

```

18. Security Rules

- All tenant-owned tables use organization_id + RLS.
- Derive tenant identity from authenticated session; do not trust browser-supplied organization_id.
- Never expose Supabase service-role key, Groq key, Google credentials, MCP token or WhatsApp bridge token to the client.
- Use least-privilege roles: owner, admin, manager, analyst, sales.
- Use audit logs for connection and important lead/admin changes.
- Encrypt/protect connector credentials.
- Support customer offboarding and data deletion.
- Rate-limit sensitive endpoints.
- Do not bypass access controls, CAPTCHA or MFA.

19. Job System

```

initial_sync
sync_social_account
sync_content
sync_comments
sync_metrics
sync_google_ads
sync_whatsapp
process_ai_comments
process_ai_conversations
generate_daily_insight
generate_weekly_insight
generate_monthly_report
calculate_attribution

```

Jobs must be idempotent. Use external IDs and unique constraints to prevent duplicates.

20. Repository Structure

```

skyletic/
├── app/
│   ├── (auth)/
│   ├── dashboard/
│   ├── social/
│   └── ads/

```

```

|   ├── whatsapp/
|   ├── leads/
|   ├── bookings/
|   ├── insights/
|   ├── reports/
|   ├── connections/
|   └── settings/
|   ├── components/
|   └── lib/
|       ├── db/ auth/ analytics/ ai/ leads/ attribution/
|       ├── connectors/
|       │   ├── instagram/
|       │   ├── facebook/
|       │   ├── youtube/
|       │   ├── google-ads/
|       │   └── whatsapp/
|       └── validation/
|   ├── workers/
|   ├── supabase/migrations/
|   ├── docs/
|   ├── tests/
|   ├── .env.example
|   ├── CLAUDE.md
|   ├── AGENTS.md
|   └── README.md

```

21. Environment Variables

```

NEXT_PUBLIC_SUPABASE_URL=
NEXT_PUBLIC_SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY=
GROQ_API_KEY=
PLAYWRIGHT_MCP_URL=
PLAYWRIGHT_MCP_TOKEN=
WHATSAPP_BRIDGE_URL=
WHATSAPP_BRIDGE_TOKEN=
GOOGLE_CLIENT_ID=
GOOGLE_CLIENT_SECRET=
GOOGLE_ADS_DEVELOPER_TOKEN=
APP_BASE_URL=
ENCRYPTION_KEY=

```

Real secrets must never be committed or exposed through NEXT_PUBLIC variables.

22. UI Requirements

- Clean professional enterprise SaaS design.
- Responsive desktop/tablet/mobile.
- Sidebar: Overview, Analytics, Social, Ads, WhatsApp, Leads, Bookings, AI Insights, Reports, Connections, Settings.
- Every KPI supports date filtering and period comparison.
- AI cards show reporting period and evidence.
- Clear loading, empty, error and disconnected states.
- Filters: platform, campaign, content, date, lead status, owner.
- Clearly distinguish verified metrics from attributed/estimated values.

23. Antigravity Master Instruction

You are the lead engineer building Skyletic, a production-minded multi-tenant SaaS platform for AI marketing, lead and conversion intelligence.

PRODUCT:

Skyletic connects Instagram, Facebook, YouTube, Google Ads, WhatsApp and a booking calendar. It normalizes marketing and customer interaction data and provides daily, weekly and monthly AI-powered business insights.

STACK:

Next.js + TypeScript; Vercel; Supabase PostgreSQL/Auth/RLS; Render workers; Playwright MCP/Playwright; Groq; existing Baileys bridge; Tailwind/shadcn; Zod.

RULES:

1. Multi-tenant from day one.
2. organization_id + RLS on tenant data.
3. Secrets server-side only.
4. Connectors behind interfaces.
5. Browser automation is a connector, not business logic.
6. Prefer official APIs/OAuth when appropriate and permitted.
7. Sync jobs must be idempotent.
8. Calculate authoritative metrics in SQL/TypeScript, not LLM.
9. Groq explains verified evidence.
10. Validate AI JSON with Zod.
11. Store AI evidence, model and prompt version.
12. Never invent facts.
13. Every DB change is a migration.
14. Test tenant isolation and critical workflows.
15. Explain plan before major implementation.
16. Implement one milestone at a time.
17. Do not rewrite working code unnecessarily.

CORE FLOW:

Social/Ads → Content/Interaction → Lead → WhatsApp → Booking → Conversion.

INITIAL TASK:

Inspect the repository only. Return:

- current structure
- technologies
- missing infrastructure
- proposed DB schema
- connector interfaces
- security risks
- implementation milestones
- files to create/change

Do not implement major features until the plan is clear.

24. Antigravity Prompt: Database Foundation

Implement Milestone 1 for Skyletic.

Create Supabase migrations for:

organizations, organization_members, social_accounts, content, content_metrics, comments, leads, lead_events, ad_accounts, ad_campaigns, ad_metrics, whatsapp_conversations, whatsapp_messages, bookings, ai_insights, sync_jobs and audit_logs.

Requirements:

- UUID keys
- organization_id on tenant tables
- foreign keys and timestamps
- unique external IDs scoped correctly
- RLS policies
- roles owner/admin/manager/analyst/sales
- useful indexes
- tenant-isolation tests
- migrations only

Explain the migration/security plan before coding.

25. Antigravity Prompt: Playwright MCP Foundation

Implement the Skyletic browser connector foundation.

Create:

- connector interface
- browser session abstraction
- Playwright MCP client/service abstraction
- secure server-side configuration
- connection status
- sync-job integration
- retry/error handling
- audit logging
- connector health checks

Rules:

- MCP credentials never reach frontend.
- Render is the runtime for long-running browser work.
- Initial MVP supports isolated account/session operation.
- CAPTCHA/MFA becomes human-action-required.
- Platform-specific selectors stay inside connector modules.
- Return normalized data.
- Add mocked connector tests.
- Do not implement platform scraping selectors yet.

26. Antigravity Prompt: WhatsApp Bridge

Integrate Skyletic with the existing Baileys bridge.

Implement:

- health check
- conversation sync
- message ingestion
- contact matching
- lead creation/update
- lead status
- unread counts
- last activity
- follow-up flag
- AI conversation summary job
- webhook authentication
- idempotency by bridge message ID

Do not rewrite the Baileys bridge. Use a clean adapter. Keep all credentials server-side.

27. Antigravity Prompt: AI Intelligence

Implement Skyletic AI using Groq.

Create:

- AI provider interface
- Groq provider
- Zod schemas
- versioned prompts
- comment analyzer
- conversation analyzer
- lead scorer
- daily insight generator
- weekly insight generator
- monthly report generator

Rules:

- metrics come from application analytics
- AI receives evidence packets
- no invented facts
- structured JSON for machine workflows
- Zod validation
- save model/prompt version/period/evidence

Start with comment classification and lead scoring.

28. Antigravity Prompt: Dashboard

Build the first production-quality Skyletic Overview.

Show:

views/reach, engagement, new leads, qualified leads, WhatsApp conversations, bookings, ad spend, CPL, conversion rate, platform comparison, top content, lead funnel, latest AI insight, recommendations and alerts.

All numbers must come from database analytics services.
Do not hardcode fake production numbers.
Implement loading, empty, error and disconnected states.

29. 30-Day Build Plan

Days 1–3:

Repo, AGENTS.md/CLAUDE.md, docs, Vercel/Supabase/Render.

Days 4–6:

Auth, organization, roles and RLS.

Days 7–9:

Core migrations and seed/test data.

Days 10–12:

Dashboard shell, navigation and KPI framework.

Days 13–15:

Connector interfaces and first social connector.

Days 16–18:

Content/metrics ingestion and analytics.

Days 19–21:

Baileys integration, conversations and leads.

Days 22–23:

Booking calendar and lead-booking association.

Days 24–25:

Groq classification, lead scoring and summaries.

Days 26–27:

Daily/weekly insights.

Days 28–29:

Google Ads integration/import and attribution.

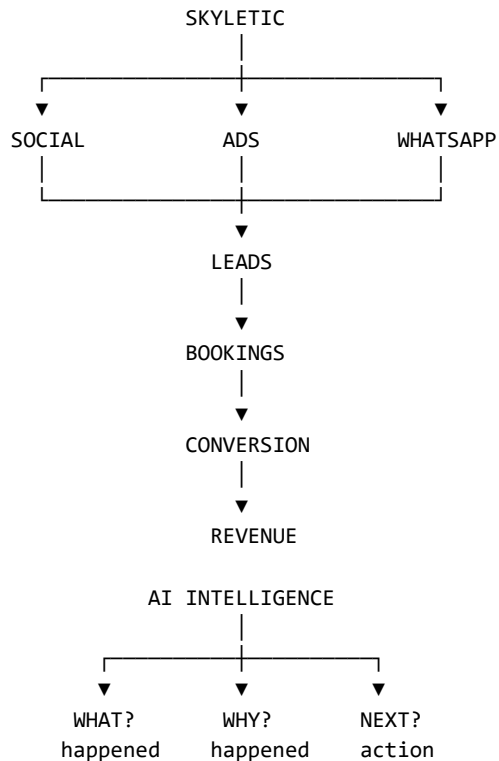
Day 30:

End-to-end tests, security review and production hardening.

30. Definition of Done

- Secure multi-tenant authentication and RLS.
- At least one social connector works end-to-end.
- Content and metrics are normalized.
- Google Ads has a production integration path.
- Baileys conversations appear in Skyletic.
- Leads and statuses work.
- Bookings associate with leads.
- Groq classifies intent and lead priority with validated JSON.
- Daily and weekly insights use verified evidence.
- Attribution connects touchpoints to leads/bookings where evidence exists.
- Render handles long-running jobs/browser workloads.
- Secrets remain server-side.
- Sync is idempotent.
- Critical workflows have tests.
- Core business logic is independent of scraping implementation.

31. Final Vision



Skyletic's differentiator is the intelligence layer that combines marketing performance, customer conversations, lead intent and bookings into one evidence-backed view of business growth.

32. Production / Compliance Notes

- Verify each platform's current terms, API policies and permitted automation/data use before commercial deployment.
- Use official APIs/OAuth where practical and permitted.
- Do not bypass CAPTCHA, MFA, access controls or platform security mechanisms.
- Obtain appropriate authorization for connected accounts.
- Protect personal and conversation data with suitable access control, retention and deletion policies.
- Document what data is collected, why, where it is stored and retention period.
- Evaluate official WhatsApp Business Platform/Cloud API before commercial scale.
- Perform a security review before customer-managed sensitive credentials are enabled.